CHENLU YE

Computer Science, University of Illinois Urbana-Champaign

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RESEARCH INTERESTS

- Self-improving agents and auto-research;
- Reinforcement learning for agentic LLM post-training;
- Decision-making problems;

EDUCATION

University of Illinois Urbana-Champaign — Urbana, USA
Ph.D. candidate, *Computer Science* — 2024.08 – Present
Advisor: Prof. Tong Zhang

The Hong Kong University of Science and Technology — Hong Kong, China
MPhil, *Artificial Intelligence* — 2021.09 – 2024.08
Advisor: Prof. Tong Zhang

University of Science and Technology of China — Hefei, China
Bachelor of Science, *Statistics* — 2017.09 – 2021.06

RESEARCH EXPERIENCE

Research Intern, Meta MSL / FAIR — 2026.05 – Present
Managers: Sainbayar Sukhbaatar and Jason Weston

- Built a **multi-agent collaboration system** to generate synthetic data for agentic tasks involving user interaction and system-level safety constraints.
- Developed a **meta-agent that accumulates skills** to improve data-generation efficiency and task quality.
- **Currently investigating** RL generalization and continual improvement for automated research, alongside asynchronous RL training stability on long-horizon, multi-turn agentic tasks.

Applied Scientist Intern, Amazon — 2025.05 – 2026.05
Hosts: Dr. Zhou Yu, Dr. Ziji Zhang

- Led the development of **Adaptive Layerwise Perturbation (ALP)** to improve off-policy RL robustness to system noise and training–inference mismatch through layerwise policy perturbations. [Paper]
- Demonstrated improvements in training stability, exploration, and task performance on mathematical reasoning and multi-turn tool-integrated reasoning tasks.
- Proposed **Process Consistency Filter (PROF)**, using process–outcome consistency for RL data selection while controlling the correct-to-incorrect response ratio. [Paper]
- Developed PROF-GRPO to improve final-answer accuracy and intermediate reasoning quality; conducted ablations on separately filtering correct and incorrect responses.

RESEARCH EXPERIENCE – CONTINUED

Ph.D. Student, University of Illinois Urbana-Champaign 2024.09 – Present

Advisor: Prof. Tong Zhang

- Developed **Reinforce-Ada**, an adaptive sampling framework that allocates generation budgets across prompts during online RL post-training to preserve informative training signals and increase their diversity. [Paper] [Code]
- Proposed a **self-rewarding correction** framework to improve model self-verification and correction in mathematical reasoning. [Paper]
- Proposed an **online iterative RLHF** framework, implemented it with online DPO, and extended it to general preference models. [Paper] [Code]

MPhil Student, The Hong Kong University of Science and Technology 2021.09 – 2024.08

Advisor: Prof. Tong Zhang

- Formulated RLHF as a reverse-KL-regularized contextual bandit problem under Bradley–Terry preferences and developed statistically efficient algorithms with finite-sample guarantees.
- Developed uncertainty-weighted algorithms for corruption-robust reinforcement learning across online and offline, value-based and model-based settings.

Visiting Research Scholar, University of California, Los Angeles 2023.08 – 2023.12

Host: Prof. Quanquan Gu

- Proposed reinforcement learning algorithms robust to adversarial corruption in online and offline decision-making.

SELECTED PUBLICATIONS AND PREPRINTS

(* denotes equal contribution)

[1] **Adaptive Layerwise Perturbation: Unifying Off-Policy Corrections for LLM RL**
Chenlu Ye*, Xuanchang Zhang*, Yifan Hao*, Zhou Yu, Ziji Zhang, Abhinav Gullapalli, Hao Chen, Jing Huang, Tong Zhang. [Preprint]

[2] **Reinforce-Ada: An Adaptive Sampling Framework under Non-linear RL Objectives**
Wei Xiong*, **Chenlu Ye***, Baohao Liao*, Hanze Dong*, Xinxing Xu, Christof Monz, Jiang Bian, Nan Jiang, Tong Zhang. [Preprint]

[3] **Beyond Correctness: Harmonizing Process and Outcome Rewards through RL Training**
Chenlu Ye, Zhou Yu, Ziji Zhang, Hao Chen, Narayanan Sadagopan, Jing Huang, Tong Zhang, Anurag Beniwal. [Preprint]

[4] **Self-rewarding correction for mathematical reasoning**
Wei Xiong*, Hanning Zhang*, **Chenlu Ye***, Lichang Chen, Nan Jiang, Tong Zhang. [Preprint]

[5] **Online iterative reinforcement learning from human feedback with general preference model**
Chenlu Ye*, Wei Xiong*, Yuheng Zhang*, Hanze Dong*, Nan Jiang, Tong Zhang. [NeurIPS 2024]

[6] **Iterative preference learning from human feedback: Bridging theory and practice for RLHF under KL-Constraint**
Wei Xiong*, Hanze Dong*, **Chenlu Ye***, Ziqi Wang, Han Zhong, Heng Ji, Nan Jiang, Tong Zhang. [ICML 2024]

[7] **Logarithmic Regret for Online KL-Regularized Reinforcement Learning**
Heyang Zhao*, **Chenlu Ye***, Wei Xiong, Quanquan Gu, Tong Zhang. [ICML 2025]

SELECTED PUBLICATIONS AND PREPRINTS – CONTINUED

[8] **Catoni Contextual Bandits are Robust to Heavy-tailed Rewards**
Chenlu Ye, Yujia Jin, Alekh Agarwal, Tong Zhang. [ICML 2025 Spotlight]

[9] **Understanding Overadaptation in Supervised Fine-Tuning: The Role of Ensemble Methods**
Yifan Hao*, Xingyuan Pan*, Hanning Zhang*, **Chenlu Ye**, Rui Pan, Tong Zhang. [ICML 2025]

[10] **Sharp Analysis for KL-Regularized Contextual Bandits and RLHF**
Heyang Zhao, **Chenlu Ye**, Quanquan Gu, Tong Zhang. [NeurIPS 2025]

[11] **Towards robust model-based reinforcement learning against adversarial corruption**
Chenlu Ye*, Jiafan He*, Quanquan Gu, Tong Zhang. [ICML 2024]

[12] **Corruption-Robust Offline Reinforcement Learning with General Function Approximation**
Chenlu Ye*, Rui Yang*, Quanquan Gu and Tong Zhang. [NeurIPS 2023]

[13] **Optimal Sample Selection Through Uncertainty Estimation and Its Application in Deep Learning**
Yong Lin*, Chen Liu*, **Chenlu Ye***, Qing Lian, Yuan Yao and Tong Zhang. [JMLR 2025]

[14] **Corruption-Robust Algorithms with Uncertainty Weighting for Nonlinear Contextual Bandits and Markov Decision Processes**
Chenlu Ye, Wei Xiong, Quanquan Gu, and Tong Zhang. [ICML 2023]

[15] **Provably Efficient High-Dimensional Bandit Learning with Batched Feedbacks**
Jianqing Fan, Zhaoran Wang, Zhuoran Yang, **Chenlu Ye** *(alphabetical author order)*. [Preprint]

[16] **Daunce: Data Attribution through Uncertainty Estimation**
Xingyuan Pan*, **Chenlu Ye***, Joseph Melkonian*, Jiaqi W. Ma, Tong Zhang. [Preprint]

[17] **Transformers as Multi-task Learners: Decoupling Features in Hidden Markov Models**
Yifan Hao*, **Chenlu Ye***, Chi Han, Tong Zhang. [Preprint]

SKILLS

Programming and ML: Python, PyTorch, C++
Developer Tools: Git, Docker

HONORS AND AWARDS

Gold Prize for Outstanding Student Scholarship (1/40) 2020.09
Bronze Prize for Outstanding Student Scholarship 2019.09
Bronze Prize for Outstanding Student Scholarship 2018.09

PROFESSIONAL SERVICE

Conference Reviewer: ICML, NeurIPS, ICLR, AISTATS.

Journal Reviewer: JMLR, Machine Learning, Artificial Intelligence.